MIL-C-52829A(ME)
16 August 1982
SUPERSEDING
MIL-C-52829(ME)
22 July 1974

MILITARY SPECIFICATION

CARRIER, TRAMWAY OR CABLEWAY

This specification is approved for use by the Mobility Equipment Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of carrier for an aerial tramway or cableway.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5420

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SPECIFICATIONS

FEDERAL

FF-B-575	- Bolts, Hexagon and Square.
FF-N-836	- Nut: Square, Hexagon, Cap, Slotted, Castle, Knurled, Welding and Single Ball Seat.
QQ-S-741	- Steel, Carbon: Structural Shapes; Plates; and Bars.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	- Boxes, Shipping, Fiberboard.

MILITARY

MIL-P-116	- Preservation, Methods of.
MIL-T-704	- Treatment and Painting of Materiel.

STANDARDS

FEDERAL

FED. STD. H28/2	- Screw-thread Standards for Federal Services, Section 2, Unified Thread Form and Thread Series for Bolts, Screws, Nuts, Tapped Holes and General Applications.
FED. STD. No. 595	- Colors.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.
MS35338	- Washer, Lock-Spring, Helical, Regular (Medium) Series, (in/mm).

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should by obtained from the contracting activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/ASME - Boiler and Pressure Vessel Code, Section IX, Welding Qualifications

ANSI/ASTM A 108 - Steel Bars, Carbon, Cold-Finished, Standard Quality.

ANSI/ASTM A 366 - Steel, Carbon, Cold-Rolled Sheet, Commercial Quality.

ANSI/ASTM A 668 - Steel Forgings, Carbon and Alloy, for General Industrial Use.

ANSI/AWS D1.1 - Structural Welding Code, Section 5, Qualification.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018).

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The carrier shall be as shown in figures 1 through 4 and as specified herein. The carrier consists of two identical sections which may be bolted together to form a complete carrier assembly as shown in figure 1, or each section can be used by itself to form a short carrier assembly as shown in figure 2.

3.2 Material. Material shall be as specified herein. However, when a definite material is not specified, a material shall be used which will enable the carrier to meet the performance requirements of this specification. Acceptance or approval of any constituent material shall not be construed as a guaranty of the acceptance of the finished product (see 6.3 and 6.4). Dissimilar metals as defined in MIL-T-704 shall not be used in intimate contact with each other unless the metals are protected against galvanic corrosion.

3.2.1 Steel sheet. The bottom shall be steel conforming to ANSI/ASTM A 366 cold-rolled sheet, commercial quality, class 1, matte finish.

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3.2.2 Angles. Angles shall be steel conforming to QQ-S-741.

3.2.3 Carbon steel. Carbon steel shall conform to ANSI/ASTM A 108, UNS G 10180 or UNS 10200. Rods shall be furnished in round form and bars shall be furnished in flat form. Plates shall be fabricated from flat stock.

3.2.4 Shackles. Shackles shall be steel conforming to ANSI/ASTM A 668.

3.3 Bolts. Bolts shall conform to FF-B-575, type II, not less than grade 5.

3.4 Nuts. Nuts shall conform to FF-N-836, type II, style 13.

3.5 Screw threads. All screw threads shall conform to FED. STD. H28/2.

3.6 Lockwashers. Lockwashers with 0.375 inside diameter (ID) shall conform to MS35338-141. Lockwashers with 0.500 ID shall conform to MS35338-143.

3.7 Treatment and painting. The portions of the carrier normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704, type A, except that the color shall conform to FED. STD. No. 595, color No. 14064. Bolts and nuts shall not be painted.

3.8 Workmanship.

3.8.1 Castings and forgings. All parts, components, and assemblies of the carrier which include castings and forgings shall be clean from harmful extraneous material such as sand, dirt, sprues, scale, and flux. Rework shall be limited to procedures which do not reduce strength or affect function.

3.8.2 Metal fabrication. Metal used in fabrication shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause damage to the material. Corners shall be square and true. Flame-cutting, using tips suitable for the thickness of the steel, may be employed instead of shearing and sawing. All bends shall be made with controlled means to insure uniformity of size and shape. Precaution shall be taken to avoid overheating. Heated steel shall be allowed to cool slowly. External surfaces shall be free from burrs, sharp edges and corners, except when sharp edges or corners are required or where they are not detrimental to safety.

3.8.3 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, mill scale that can be removed by chipping and wire brushing, and other foreign matter. Parent materials, weld filler metals, and fabrication techniques shall be as required to enable the carrier to conform to the examination requirement specified in 4.4.1. Parts to be joined by fillet welds shall be brought into as close contact as possible and in no event shall be separated by more than 0.1875 inch, unless appropriate bridging techniques are used. Unless otherwise specified (see 6.2), the welding process used in fabrication of the carrier shall be at the option of the contractor.

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3.8.4 Welders. Before assigning any welder to manual welding work covered by this specification, the contractor shall provide the contracting officer with certification that the welder has passed qualification tests as prescribed by either of the following listed codes for the type of welding operations to be performed and that such qualification is effective as defined by the particular code.

ANSI/AWS D1.1, Structural Welding Code, Section 5, Qualification.

ANSI/ASME Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

Contractors who make only horizontal welds need not qualify welders for "all position welding". Subject to approval by the Government, the contractor's standard welder qualification may be substituted in lieu of the above codes provided that the contractor's procedure is equivalent to the above codes. The contractor shall be responsible for determining that automatic welding equipment operators are capable of producing quality welds in accordance with the AWS or ASME codes.

3.8.5 Welding practices.

a. Preheat of materials being welded, and maximum interpass temperature during welding, shall be in accordance with the contractor's recommendations.

b. Tack welds shall be subject to the same quality requirements as final welds.

c. Work shall be positioned for flat welding whenever practicable.

d. Procedures and sequences shall be such that distortion and shrinkage will be held to a minimum. When straightening is required, caution shall be exercised to insure that the straightening process does not weaken the part and carrier.

3.8.6 Weld quality.

a. The contractor shall be responsible for determining the inspections needed to insure that, when the welded parts are assembled together to make the carrier, the carrier shall conform to the inspection requirements specified in section 4.

b. All weldments shall be free from slag, flux, weld spatter, and other impurities detrimental to both the appearance and strength of the weldment.

c. Undercut in weldments shall be held to a minimum but shall be not more than 0.0312 inch deep.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Quality conformance inspection (see 4.3).

b. Inspection of packaging (see 4.5).

4.3 Quality conformance inspection.

4.3.1 Sampling. Sampling for examination shall be in accordance with MIL-STD- inspection level III.

4.3.2 Examination. Samples selected in accordance with 4.3.1 shall be examined for the defects specified in 4.4.1 AQL shall be 0.4 percent defective.

4.4 Inspection procedure.

4.4.1 Examination. The carriers shall be examined as specified herein for the following defects:

101. Material not as specified.
102. Carrier not as shown in figure 1 through 4.
103. Treatment and painting not as specified.
104. Workmanship not as specified.

4.5 Inspection of packaging.

4.5.1 Quality conformance inspection of pack.

4.5.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

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4.5.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.5.1.3 Examination. Samples selected in accordance with 4.5.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

105.	Unprotected surfaces not coated with preservative as specified for level A.
106.	Small loose components not contained together as specified for level A.
107.	Boxes not as specified for level A or B.
108.	Components not blocked and braced within the box as specified for level A or B.
109.	Packaging not in accordance with the referenced document as specified for commercial.
110.	Marking missing, illegible, incorrect, or incomplete for level A, B, or commercial.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial as specified (see 6.2).

5.1.1 Level A. The carrier shall be disassembled into a various assemblies and components. Unpainted metal surfaces requiring the application of a contact preservative in accordance with MIL-P-116 shall be coated with type P-1 preservative. The preservative shall conform to the applicable specification listed in and shall be applied in accordance with MIL-P-116. Loose components such as bolts, nuts, washers, shackles, or any other small loose components, requiring protection to prevent damage or loss shall be contained together in a close-fitting box conforming to PPP-B-636, W5c or W6c as applicable to the weight of the contents. Box closure shall be in accordance with the appendix to the box specification.

5.1.2 Commercial. The components comprising a complete carrier shall be preserved in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be level A, B, or commercial as specified (see 6.2).

5.2.1 Level A. Each complete carrier, preserved as specified in 5.1, shall be packed in a close-fitting box conforming to PPP-B-601, overseas type, style optional; or PPP-B-621, class 2, style optional. The contents shall be blocked and braced within the box in a manner to prevent free movement or damage. Box closure shall be in accordance with the appendix to the applicable box specification.

5.2.2 Level B. The carriers shall be packed as specified in 5.2.1 for level A, except the boxes shall be domestic type or class as applicable.

5.2.3 Commercial. The carriers preserved as specified in 5.1 shall be packed in accordance with MIL-STD-1188.

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5.3 Marking.

5.3.1 Military. Marking for military levels of protection (level A or B) shall be in accordance with MIL-STD-129

5.3.2 Commercial. Marking for commercial packaging shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The carrier is intended to be used as the load-supporting platform upon which materiel is transported by means of the Light, Pioneer M-1, Aerial Tramway and Cableway.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Welding process required when other than at the option of the contractor, (see 3.8.3).

c. Degree of preservation and degree of packing required (see 5.1 and 5.2).

6.3 Recycled material. It is encouraged that recycled material be used, when practical, as long as it meets the requirements of this specification (seee 3.2).

6.4 Materiel deterioration prevention and control (MADPAC). The contracting officer should consider for inclusion in all applicable invitations for bid, requests for proposal, and contracts the following statement:

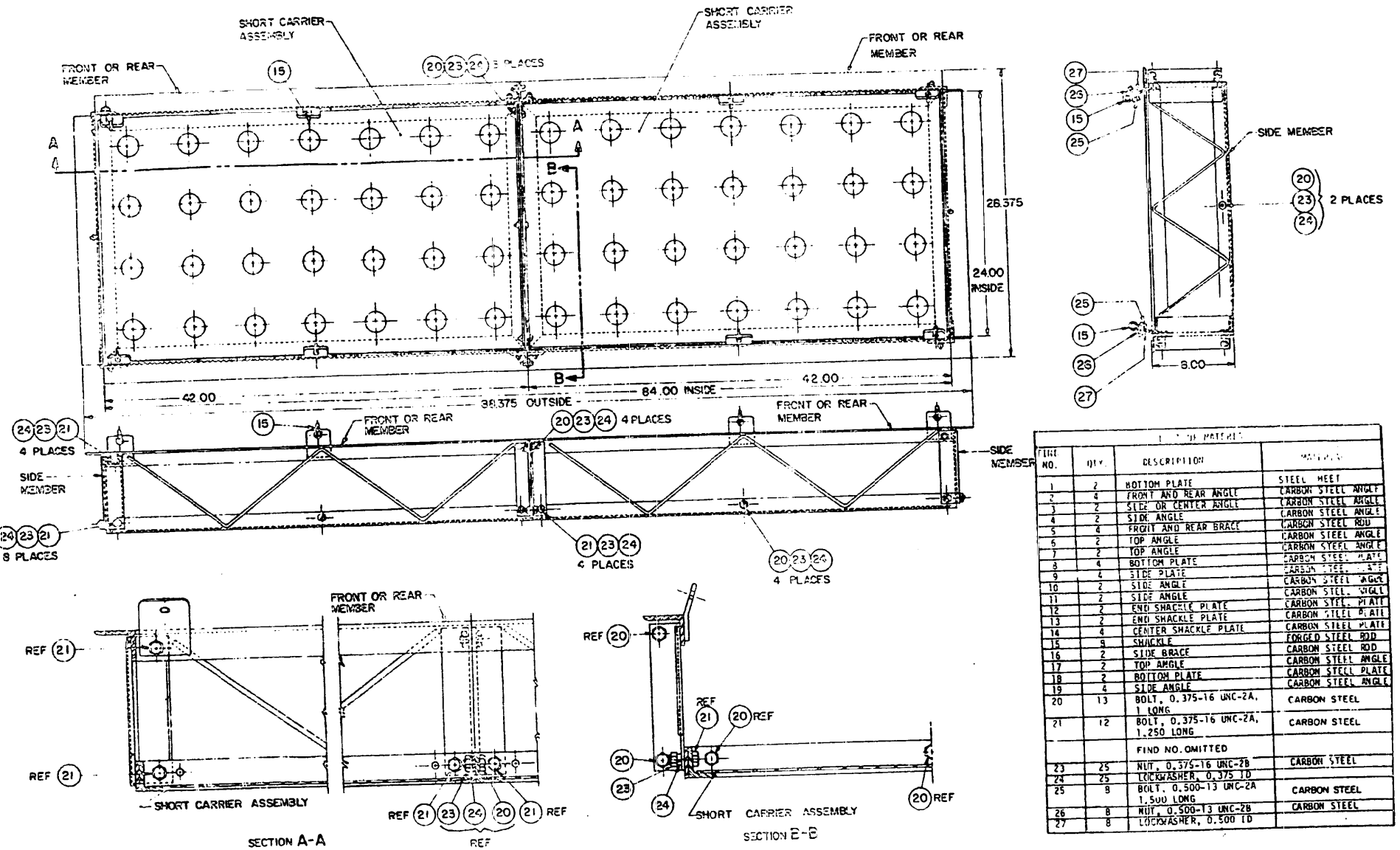
"The technical proposal shall provide assurance that all system/item parts shall be fabricated from compatible materials, suitably finished, to provide protection from corrosion and other forms of material deterioration. The assurance shall be in the form of a finish plan, utilizing state-of-art technology, which clearly identifies the specific finish or treatment to be used on the various components and assemblies, to provide protection against the various forms of materiel deterioration in the applicable operating and storage environments to which they may be exposed." (see 3.2).

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ITEM NO.	QTY.	DESCRIPTION	MATERIAL
1	2	BOTTOM PLATE	STEEL SHEET
2	4	FRONT AND REAR ANGLE	CARBON STEEL ANGLE
3	2	SIDE OR CENTER ANGLE	CARBON STEEL ANGLE
4	2	SIDE ANGLE	CARBON STEEL ANGLE
5	4	FRONT AND REAR BRACE	CARBON STEEL ROD
6	2	TOP ANGLE	CARBON STEEL ANGLE
7	2	TOP ANGLE	CARBON STEEL ANGLE
8	4	BOTTOM PLATE	CARBON STEEL PLATE
9	4	SIDE PLATE	CARBON STEEL PLATE
10	2	SIDE ANGLE	CARBON STEEL ANGLE
11	2	SIDE ANGLE	CARBON STEEL ANGLE
12	2	END SHACKLE PLATE	CARBON STEEL PLATE
13	2	END SHACKLE PLATE	CARBON STEEL PLATE
14	4	CENTER SHACKLE PLATE	CARBON STEEL PLATE
15	9	SHACKLE	FORGED STEEL ROD
16	2	SIDE BRACE	CARBON STEEL ROD
17	2	TOP ANGLE	CARBON STEEL ANGLE
18	2	BOTTOM PLATE	CARBON STEEL PLATE
19	4	SIDE ANGLE	CARBON STEEL ANGLE
20	13	BOLT, 0.375-16 UNC-2A, 1 LONG	CARBON STEEL
21	12	BOLT, 0.375-16 UNC-2A, 1.250 LONG	CARBON STEEL
		FIND NO. OMITTED	
23	25	NUT, 0.375-16 UNC-2B	CARBON STEEL
24	25	LOCKWASHER, 0.375 ID	
25	8	BOLT, 0.500-13 UNC-2A 1.500 LONG	CARBON STEEL
26	8	NUT, 0.500-13 UNC-2B	CARBON STEEL
27	8	LOCKWASHER, 0.500 ID	

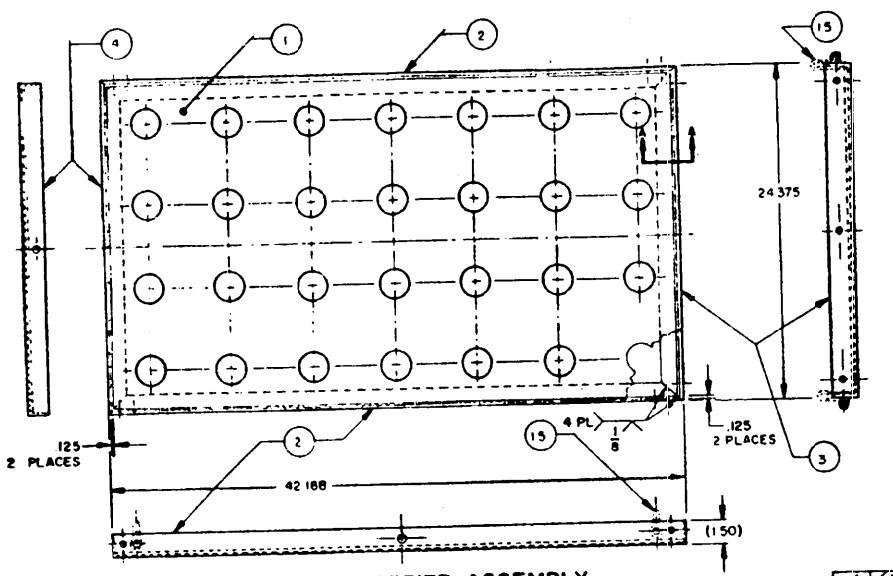
NOTES:
1 ALL DIMENSIONS ARE IN INCHES
2 CAN BE ASSEMBLED AS SHOWN ABOVE OR SECTIONS CAN BE USED INDIVIDUALLY AS SHOWN ON FIGURE 2.
3 TOLERANCES: .xx ± .06 xxx ± .030 ∠ ± 2°

FIGURE 1. Complete carrier assembly for cableway or tramway.

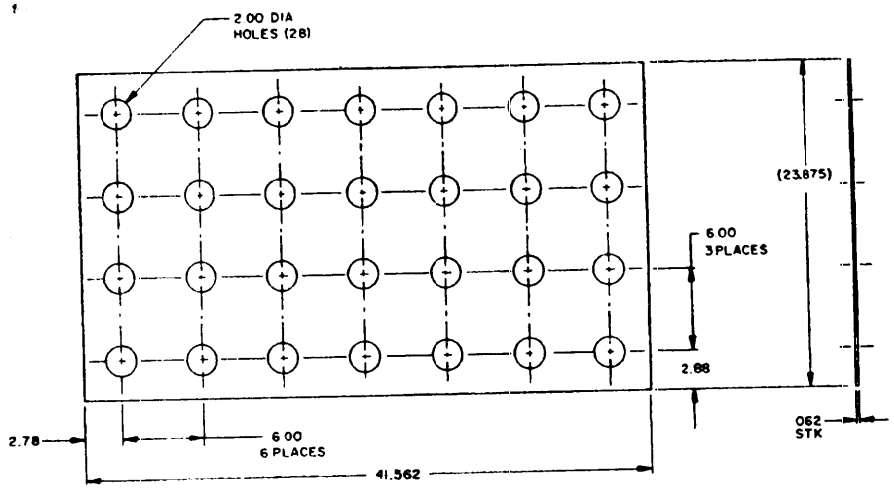
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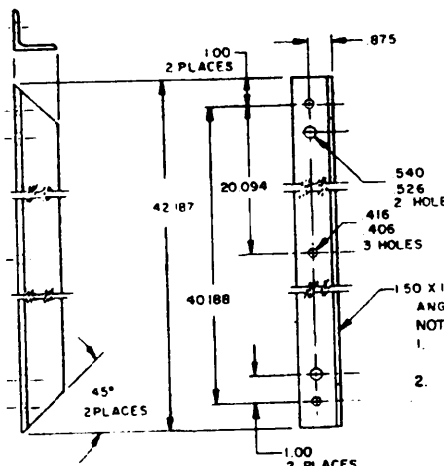
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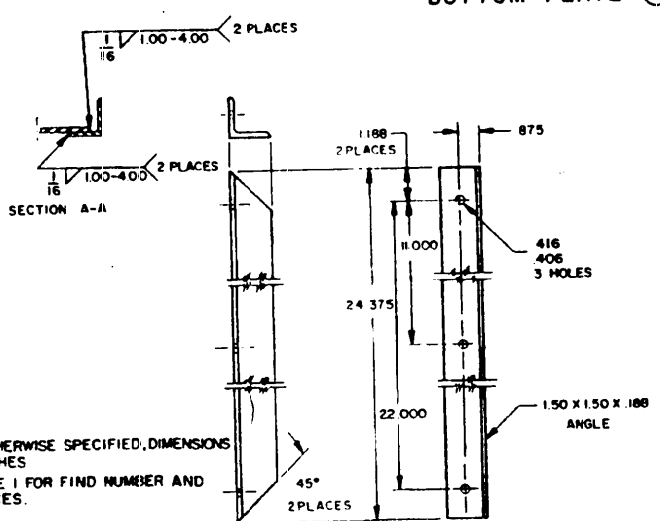
SHORT CARRIER ASSEMBLY



BOTTOM PLATE (1)

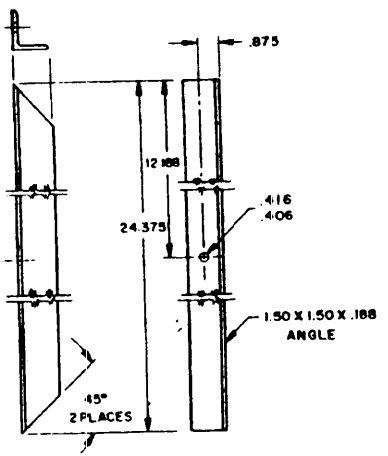


FRONT & REAR
ANGLE (2)



SIDE OR CENTER ANGLE (3)

NOTES:
1. UNLESS OTHERWISE SPECIFIED, DIMENSIONS ARE IN INCHES
2. SEE FIGURE 1 FOR FIND NUMBER AND TOLERANCES.



SIDE ANGLE (4)

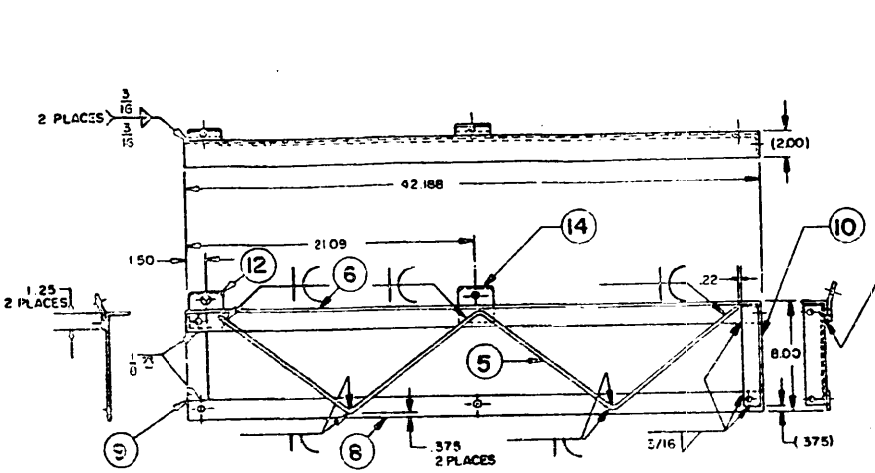
FIGURE 2. Assembly & details of short carrier.

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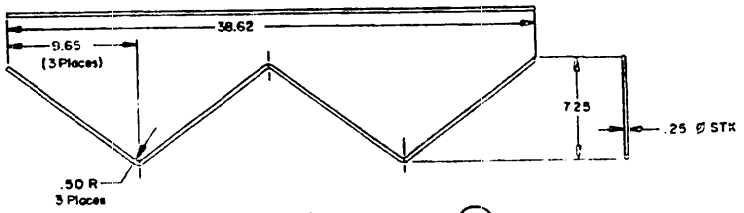
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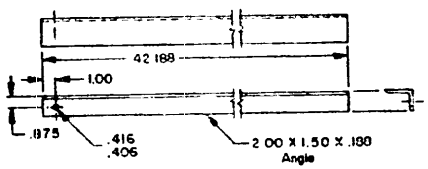


ASSEMBLY OF FRONT & REAR MEMBERS

Four (4) Reqs. 2 As Shown and 2 Opp. hand.

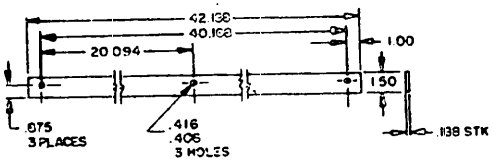


FRONT & REAR BRACE (5)

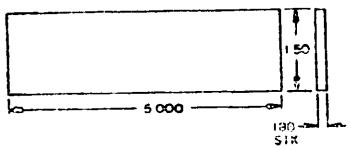


TOP ANGLE (6) (7)

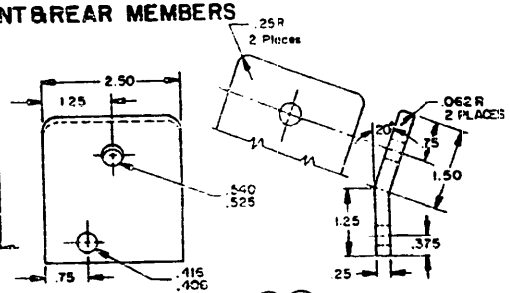
Find NO. 6 As Shown. Find NO 7 Opp. hand.



BOTTOM PLATE (6)

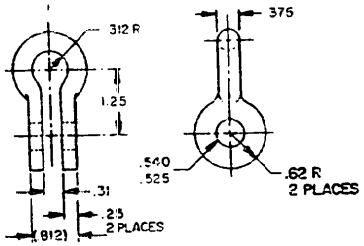


SIDE PLATE (9)

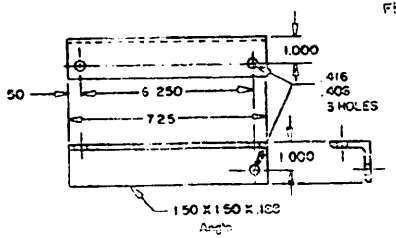


SHACKLE PLATE (12) (13)

Find NO. 12 As Shown. Find NO. 13 Opp. hand.

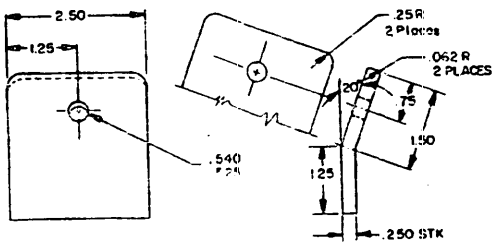


SHACKLE (15)



SIDE ANGLE (10) (11)

Find NO. 10 As Shown. Find NO. 11 Opp. hand



SHACKLE PLATE (14)

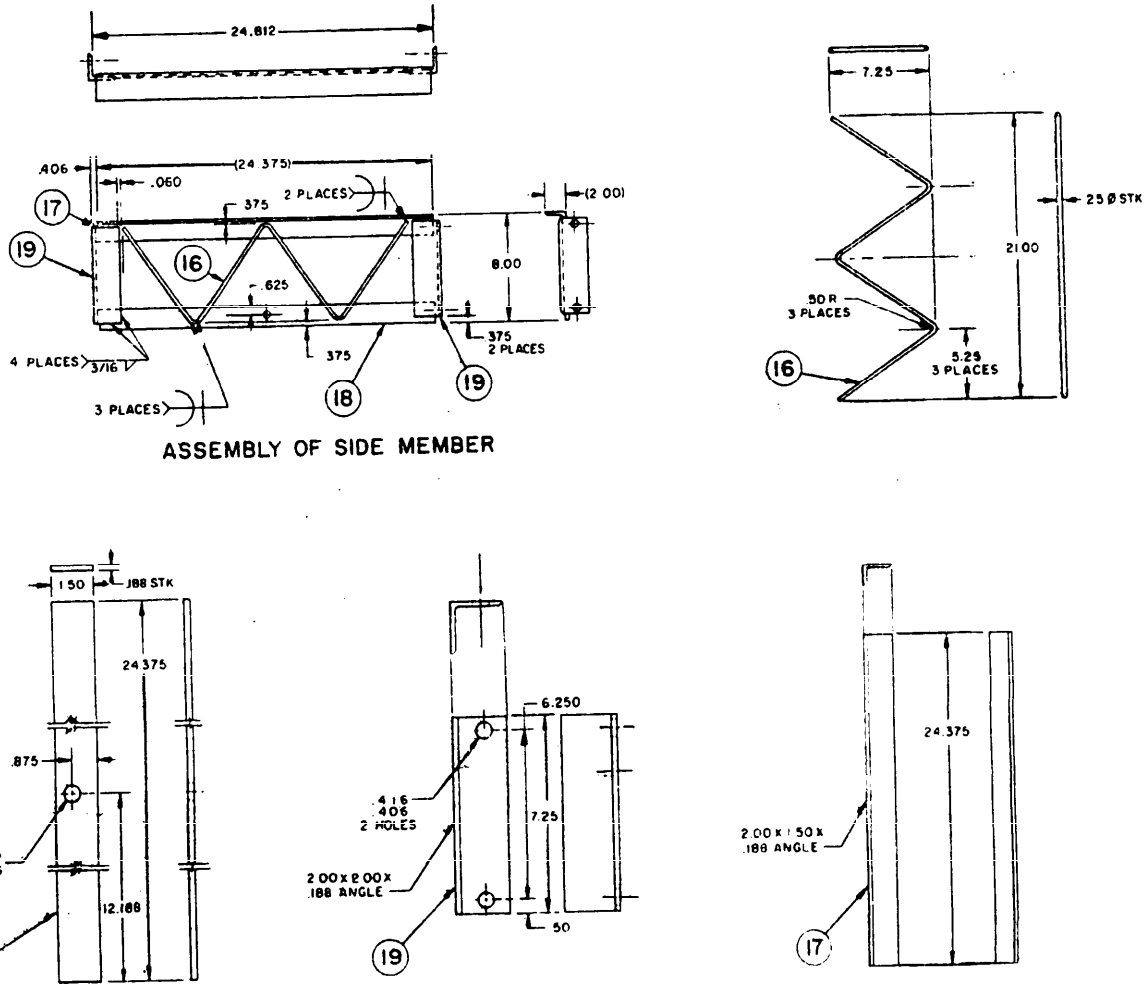
NOTES:
1. UNLESS OTHERWISE SPECIFIED, DIMENSIONS ARE IN INCHES.
2. SEE FIGURE 1 FOR FIND NUMBER AND TOLERANCES.

FIGURE 3. Assembly & details of front and rear members.

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NOTES:
1. ALL DIMENSIONS ARE IN INCHES
2. SEE FIGURE 1 FOR FIND NUMBER AND TOLERANCES.

FIGURE 4. Assembly & details of side members.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-C-52829A(ME) Carrier, Tramway or Cableway

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*

TELEPHONE NO.

DATE

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EDITION OF 1 JAN 72 WILL BE USED UNTIL EXHAUSTED.

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